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JupyterLab 🗨 🛠 Python 3 (ipykernel) ○ ≡

```
•[1]: marks = [78, 92, 65, 88, 74]
      print("Original List:", marks)
      marks.append(95)
      print("After append:", marks)
      marks.insert(2, 80)
      print("After insert:", marks)
      marks.remove(65)
      print("After remove:", marks)
      marks.sort()
      print("After sort:", marks)
      print("Final Updated List:", marks)
```

```
Original List: [78, 92, 65, 88, 74]
After append: [78, 92, 65, 88, 74, 95]
After insert: [78, 92, 80, 65, 88, 74, 95]
After remove: [78, 92, 80, 88, 74, 95]
After sort: [74, 78, 80, 88, 92, 95]
Final Updated List: [74, 78, 80, 88, 92, 95]
```

```
•[2]: cities = ("Delhi", "Mumbai", "Chennai", "Bangalore", "Kolkata", "Hyderabad")
```

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```
print("Last city:", cities[-1])
print("Total cities:", len(cities))
if "Delhi" in cities:
    print("Delhi exists in the tuple")
else:
    print("Delhi does not exist in the tuple")
```

First city: Delhi
Last city: Hyderabad
Total cities: 6
Delhi exists in the tuple

```
•[3]: A = {1, 2, 3, 4, 5}
      B = {4, 5, 6, 7}
      print("Union:", A.union(B))
      print("Intersection:", A.intersection(B))
      print("Difference (A - B):", A.difference(B))
      print("Symmetric Difference:", A.symmetric_difference(B))
```

Union: {1, 2, 3, 4, 5, 6, 7}
Intersection: {4, 5}
Difference (A - B): {1, 2, 3}
Symmetric Difference: {1, 2, 3, 6, 7}

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```
•[5]: student = {  
    "name": "Krithika",  
    "roll": 101,  
    "branch": "CSE",  
    "marks": 85  
}  
student["marks"] = 92  
print("Keys:")  
print(student.keys())  
print("\nValues:")  
print(student.values())
```

Keys:
dict_keys(['name', 'roll', 'branch', 'marks'])

Values:
dict_values(['Krithika', 101, 'CSE', 92])

```
•[6]: items = ["Pen", "Book", "Bag", "Bottle"]  
prices = [20, 150, 500, 80]  
total_bill = sum(prices)  
highest_price = max(prices)  
lowest_price = min(prices)
```

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Python 3 (ipykernel)



```
highest_price = max(prices)
lowest_price = min(prices)
print("Items:", items)
print("Prices:", prices)
print("\nTotal Bill:", total_bill)
print("Highest Price:", highest_price)
print("Lowest Price:", lowest_price)
```

Items: ['Pen', 'Book', 'Bag', 'Bottle']

Prices: [20, 150, 500, 80]

Total Bill: 750

Highest Price: 500

Lowest Price: 20

```
•[7]: def addition(a, b):
      return a + b

      def subtraction(a, b):
      return a - b

      def multiplication(a, b):
      return a * b
```

```
def division(a, b):  
    if b != 0:  
        return a / b  
    else:  
        return "Division by zero is not allowed"  
num1 = float(input("Enter first number: "))  
num2 = float(input("Enter second number: "))  
print("Addition:", addition(num1, num2))  
print("Subtraction:", subtraction(num1, num2))  
print("Multiplication:", multiplication(num1, num2))  
print("Division:", division(num1, num2))
```

```
Enter first number: 3  
Enter second number: 5  
Addition: 8.0  
Subtraction: -2.0  
Multiplication: 15.0  
Division: 0.6
```

```
[8]: def check_even_odd(n):  
      if n % 2 == 0:  
          return "Even"  
      else:
```

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```
else:
```

```
    return "Odd"
```

```
num = int(input("Enter a number: "))
```

```
print(check_even_odd(num))
```

```
Enter a number: 6
```

```
Even
```

```
[9]: def find_largest(a, b, c):
```

```
     if a >= b and a >= c:
```

```
         return a
```

```
     elif b >= a and b >= c:
```

```
         return b
```

```
     else:
```

```
         return c
```

```
num1 = int(input("Enter first number: "))
```

```
num2 = int(input("Enter second number: "))
```

```
num3 = int(input("Enter third number: "))
```

```
print("Largest number is:", find_largest(num1, num2, num3))
```

```
Enter first number: 5
```

```
Enter second number: 7
```

```
Enter third number: 8
```

```
Largest number is: 8
```

```
[10]: def factorial(n):
      fact = 1
      for i in range(1, n + 1):
          fact = fact * i
      return fact
num = int(input("Enter a number: "))
print("Factorial is:", factorial(num))
```

```
Enter a number: 9
Factorial is: 362880
```

```
•[14]: def is_prime(n):
      if n <= 1:
          return False

      for i in range(2, n):
          if n % i == 0:
              return False

      return True
num = int(input("Enter a number: "))
print(is_prime(num))
```

Enter a number: 3
True

```
•[16]: students = ["Anu", "Rahul", "Kiran"]
def add_student(name):
    students.append(name)
    print(name, "added successfully")
def remove_student(name):
    if name in students:
        students.remove(name)
        print(name, "removed successfully")
    else:
        print(name, "not found")
def display_students():
    print("Student List:")
    for student in students:
        print(student)
add_student("Priya")
remove_student("Rahul")
display_students()
```

Priya added successfully
Rahul removed successfully

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kanu removed successfully

Student List:

Anu

Kiran

Priya

```
*[18]: employees = {
    "Ravi": 25000,
    "Anu": 30000
}

def add_employee(name, salary):
    employees[name] = salary
    print(name, "added successfully")

def update_salary(name, salary):
    if name in employees:
        employees[name] = salary
        print("Salary updated for", name)
    else:
        print(name, "not found")

def search_employee(name):
```



```
def search_employee(name):
    if name in employees:
        print(name, "Salary:", employees[name])
    else:
        print(name, "not found")
add_employee("Kiran", 28000)
update_salary("Anu", 35000)
search_employee("Anu")
print("\nEmployee Records:")
print(employees)
```

```
Kiran added successfully
Salary updated for Anu
Anu Salary: 35000
```

```
Employee Records:
{'Ravi': 25000, 'Anu': 35000, 'Kiran': 28000}
```

```
*[19]: for i in range(10):
        num = int(input("Enter a number: "))
        numbers.append(num)
    unique_numbers = set(numbers)
    print("Unique numbers are:", unique_numbers)
```



```
print(Unique numbers are: , unique_numbers)
print("Count of unique numbers:", len(unique_numbers))
```

```
Enter a number: 9
Enter a number: 3
Enter a number: 8
Enter a number: 6
Enter a number: 2
Enter a number: 4
Enter a number: 1
Enter a number: 6
Enter a number: 7
Enter a number: 4
Unique numbers are: {1, 2, 3, 4, 6, 7, 8, 9}
Count of unique numbers: 8
```

```
•[20]: ages = (12, 18, 25, 16, 30, 20)
def check_voting(age):
    if age >= 18:
        return "Eligible for voting"
    else:
        return "Not eligible for voting"
for age in ages:
    print(age, "-", check_voting(age))
```

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Python 3 (ipykernel)



```
12 - Not eligible for voting
18 - Eligible for voting
25 - Eligible for voting
16 - Not eligible for voting
30 - Eligible for voting
20 - Eligible for voting
```

```
•[21]: sentence = input("Enter a sentence: ")
words = sentence.split()
```

```
frequency = {}
for word in words:
    if word in frequency:
        frequency[word] += 1
    else:
        frequency[word] = 1
print(frequency)
```

```
Enter a sentence: beautiful day
{'beautiful': 1, 'day': 1}
```

```
•[22]: books = []
def add_book():
    book = input("Enter book name: ")
```

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JupyterLab Python 3 (ipykernel)

```
<       Code   
book = input("Enter book name: ")  
books.append(book)  
print(book, "added successfully")  
def remove_book():  
    book = input("Enter book name to remove: ")  
    if book in books:  
        books.remove(book)  
        print(book, "removed successfully")  
    else:  
        print("Book not found")  
def search_book():  
    book = input("Enter book name to search: ")  
    if book in books:  
        print(book, "is available")  
    else:  
        print(book, "is not available")  
def display_books():  
    if len(books) == 0:  
        print("No books available")  
    else:  
        print("Book List:")  
        for book in books:  
            print(book)
```

```
print(book)

while True:
    print("\n--- Library Book System ---")
    print("1. Add Book")
    print("2. Remove Book")
    print("3. Search Book")
    print("4. Display Books")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        add_book()

    elif choice == 2:
        remove_book()

    elif choice == 3:
        search_book()

    elif choice == 4:
        display_books()
```



```
elif choice == 5:
    print("Exiting program...")
    break

else:
    print("Invalid choice")
```

--- Library Book System ---

1. Add Book
2. Remove Book
3. Search Book
4. Display Books
5. Exit

Enter your choice: 1
Enter book name: library book
library book added successfully

--- Library Book System ---

1. Add Book
2. Remove Book
3. Search Book
4. Display Books
5. Exit

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Python 3 (ipykernel)



Enter your choice: 5

Exiting program...

```
3]: def calculate_average(marks):  
    return sum(marks) / len(marks)  
def find_highest(marks):  
    return max(marks)  
def find_lowest(marks):  
    return min(marks)  
  
def count_passed(marks):  
    count = 0  
    for mark in marks:  
        if mark > 40:  
            count += 1  
    return count  
marks = []  
  
for i in range(5):  
    mark = int(input(f"Enter marks of student {i+1}: "))  
    marks.append(mark)  
print("\nAverage Marks:", calculate_average(marks))
```



```
print("\nAverage Marks:", calculate_average(marks))
print("Highest Marks:", find_highest(marks))
print("Lowest Marks:", find_lowest(marks))
print("Passed Students:", count_passed(marks))
```

```
Enter marks of student 1: 70
Enter marks of student 2: 80
Enter marks of student 3: 67
Enter marks of student 4: 45
Enter marks of student 5: 30
```

```
Average Marks: 58.4
Highest Marks: 80
Lowest Marks: 30
Passed Students: 4
```

```
[27]: def check_password(password):
    has_digit = False
    has_upper = False
    has_lower = False
    for char in password:
        if char.isdigit():
            has_digit = True
```



```
        has_upper = True
    elif char.islower():
        has_lower = True

    if len(password) >= 8 and has_digit and has_upper and has_lower:
        return "Strong Password"
    else:
        return "Weak Password"
pwd = input("Enter password: ")

print(check_password(pwd))
```

```
Enter password: K#ngxpkmxsl
Weak Password
```

```
•[29]: contacts = {}
def add_contact():
    name = input("Enter contact name: ")
    number = input("Enter phone number: ")
    contacts[name] = number
    print("Contact added successfully")
```



```
contacts[name] = number
print("Contact added successfully")

def search_contact():
    name = input("Enter contact name to search: ")
    if name in contacts:
        print(name, ":", contacts[name])
    else:
        print("Contact not found")

def delete_contact():
    name = input("Enter contact name to delete: ")
    if name in contacts:
        del contacts[name]
        print("Contact deleted successfully")
    else:
        print("Contact not found")

def display_contacts():
    if len(contacts) == 0:
        print("No contacts available")
    else:
        print("\nContact List:")
```



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       Code

JupyterLab   Python 3 (ipykernel)  

```
for name, number in contacts.items():
    print(name, ":", number)

while True:
    print("\n--- Contact Book Application ---")
    print("1. Add Contact")
    print("2. Search Contact")
    print("3. Delete Contact")
    print("4. Display Contacts")
    print("5. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        add_contact()

    elif choice == 2:
        search_contact()

    elif choice == 3:
        delete_contact()
```

localhost:8888/notebooks/Untitled16.ipynb

Summarize🔗🔍🌟👤⋮🐍Chat

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         Code

JupyterLab🔗🔒Python 3 (ipykernel)🔵☰

```
elif choice == 5:
    print("Exiting program...")
    break

else:
    print("Invalid choice")
```

```
--- Contact Book Application ---
1. Add Contact
2. Search Contact
3. Delete Contact
4. Display Contacts
5. Exit
Enter your choice: 1
Enter contact name: contact book
Enter phone number: 963571928
Contact added successfully

--- Contact Book Application ---
```

Contact added successfully

--- Contact Book Application ---

1. Add Contact
2. Search Contact
3. Delete Contact
4. Display Contacts
5. Exit

Enter your choice: 5

Exiting program...

```
*[30]: balance = 1000

def deposit():
    global balance
    amount = float(input("Enter amount to deposit: "))
    balance += amount
    print("Amount deposited successfully")

def withdraw():
    global balance
    amount = float(input("Enter amount to withdraw: "))
```



```
if amount <= balance:
    balance -= amount
    print("Amount withdrawn successfully")
else:
    print("Insufficient balance")

def check_balance():
    print("Current Balance:", balance)

while True:
    print("\n--- Mini Banking System ---")
    print("1. Deposit")
    print("2. Withdraw")
    print("3. Check Balance")
    print("4. Exit")

    choice = int(input("Enter your choice: "))

    if choice == 1:
        deposit()

    elif choice == 2:
```



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```
elif choice == 2:
    withdraw()

elif choice == 3:
    check_balance()

elif choice == 4:
    print("Thank you for using banking system")
    break

else:
    print("Invalid choice")
```

```
--- Mini Banking System ---
1. Deposit
2. Withdraw
3. Check Balance
4. Exit
Enter your choice: 1
Enter amount to deposit: 12000
Amount deposited successfully
```




Amount deposited successfully

--- Mini Banking System ---

1. Deposit
2. Withdraw
3. Check Balance
4. Exit

Enter your choice: 5

Invalid choice

--- Mini Banking System ---

1. Deposit
2. Withdraw
3. Check Balance
4. Exit

Enter your choice: 4

Thank you for using banking system

```
[31]: numbers = [10, 20, 30, 40, 50]
```

```
maximum = numbers[0]
```

```
for num in numbers:
```

```
    if num > maximum:
```



```
maximum = numbers[0]

for num in numbers:
    if num > maximum:
        maximum = num
minimum = numbers[0]

for num in numbers:
    if num < minimum:
        minimum = num

reversed_list = []

for i in range(len(numbers) - 1, -1, -1):
    reversed_list.append(numbers[i])
print("Maximum:", maximum)
print("Minimum:", minimum)
print("Reversed List:", reversed_list)

Maximum: 50
Minimum: 10
Reversed List: [50, 40, 30, 20, 10]
```